APPENDIX A1 · INFORMATION SETS

What each method knew at each forecast date

Method	Price history	Real dates	Place	Mortgage rate	Jobs
ARIMA, naive drift, ML models	2000 to forecast date	no	no	no	no
ARIMAX	2000 to forecast date	no	no	yes, lag 24	no
ARIMAX + jobs	2000 to forecast date	no	no	yes, lag 24	yes, lag 12
AI, named prompt	2000 to forecast date	yes	yes	yes	no
AI, blind prompt	2000 to forecast date	no	no	yes	no
AI, masked prompt	last 10 years	no	no	no	no
AI, masked, all history (Claude only)	2000 to forecast date	no	no	no	no

APPENDIX A2 · MODELS

The 14 models in one line each

Model	Family	In one line
Naive drift	Baseline	Last value, grown at the average monthly growth since 2000 (a random walk with drift).
ARIMA	Classical	Forecasts from the series' own recent values and recent surprises, after differencing once; ARIMA(1,1,2) on log price.
Holt (damped)	Classical	Exponential smoothing of the level and trend, with the trend slowly flattening out.
Prophet	Classical	Meta's curve-fitting model: one smooth trend line plus changepoints; does not start from the latest value.
Prophet + COVID & rates	Classical	Prophet with the 2020-22 boom marked as a one-off and the mortgage rate (lagged 24 months) added.
XGBoost	ML	Boosted decision trees predicting next month's growth from the last 12 months of growth.
LightGBM	ML	A faster boosted-tree method, same inputs as XGBoost.
LSTM	Deep learning	A recurrent neural network that reads the last 24 months of growth and predicts the next.
GRU	Deep learning	A simpler recurrent network, same setup as the LSTM.
Chronos (foundation model)	AI, pretrained	Amazon's pretrained time-series model, used as is: it reads the price history and forecasts, never trained on our data.
ARIMAX	Outside info	ARIMA on monthly growth plus the 12-month change in the mortgage rate, lagged 24 months.
ARIMAX + jobs	Outside info	ARIMAX plus 12-month payroll job growth in LA and Orange County, lagged 12 months.
VAR (growth + rate)	Outside info	Home growth and rate changes forecast each other (11 monthly lags).
Factor VAR (neighborhoods)	Outside info	Summarizes ~400 neighborhoods into 2 common factors and forecasts them together with metro growth.

APPENDIX A3 · ROLLING-ORIGIN TEST

Full rolling-origin results, 14 models

Model	1 mo	3 mo	6 mo	12 mo	Avg 1-12	RMSE $	RMSE rank
ARIMAX	**0.09**	**0.81**	**2.45**	**4.98**	**2.67**	**35.8K**	**1**
ARIMA	0.09	0.78	2.53	5.37	2.77	38.1K	4
ARIMAX + jobs	0.12	0.94	2.63	5.12	2.81	36.2K	2
VAR (growth + rate)	0.14	0.94	2.56	5.74	2.93	47.6K	10
LSTM	0.12	0.89	2.95	5.58	3.02	43.3K	6
GRU	0.11	0.93	2.99	5.86	3.18	41.5K	5
Naive drift	0.69	1.93	3.48	4.92	3.27	37.6K	3
Holt (damped)	0.28	1.15	3.04	6.34	3.39	44.3K	7
Chronos (foundation model)	0.69	1.61	3.44	6.13	3.56	45.3K	8
LightGBM	0.22	1.23	3.32	6.97	3.60	46.5K	9
Factor VAR (neighborhoods)	0.16	1.15	3.56	6.60	3.64	50.7K	12
XGBoost	0.25	1.25	3.40	6.82	3.65	47.8K	11
Prophet + COVID & rates	3.27	4.08	5.32	6.45	5.12	53.5K	13
Prophet	5.00	5.76	6.73	8.62	6.89	70.2K	14

Memory checks on Claude

Check (Claude only)	Result
Name the blind series	Los Angeles 16 of 18, California 17 of 18; dates exact every time
Name the masked series	Los Angeles 0 of 18; guesses: unknown 7, Seattle 6, Portland 2, San Francisco 1, Dallas 1, Denver 1
Date the masked series	median 1 month off, max 4
Masked, all history since 2000	2.47% avg error (ARIMAX 2.67%), DM p = 0.25; April 2022 forecast -4.7% (actual -3.7%)
Earlier run through Claude Code	named 1.58%, masked 2.67%; the clean API rerun changed little

APPENDIX A5 · TREND CHECK

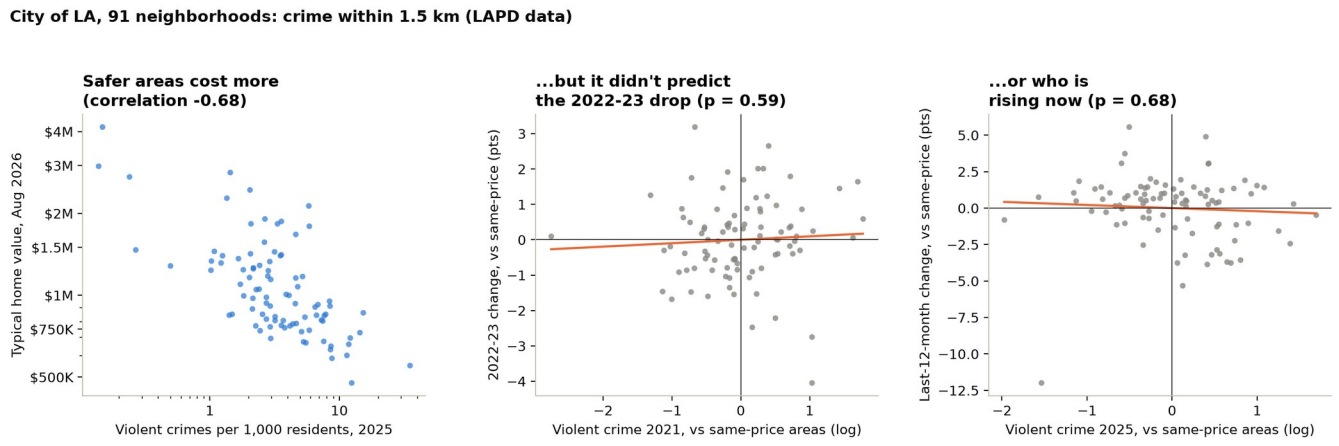
Named and blind AI tracked what happened; masked AI tracked the trend

Prompt	Claude	GPT	Gemini	Follows
Named	0.97 / 0.14	0.91 / 0.16	0.90 / 0.28	what happened
Blind	0.89 / 0.19	0.94 / 0.23	0.83 / 0.15	what happened
Masked	0.21 / 0.83	0.26 / 0.87	0.15 / 0.49	recent trend
ARIMAX	0.37 / 0.66			recent trend

NOTE Each cell: correlation of the 12-month forecast with the actual 12-month change / with the last 3 months' growth, across 18 forecast dates.

APPENDIX A6 · ONLY IF ASKED ABOUT CRIME

Crime tracks price, but it didn't predict the 2022 drop or today's winners



WHY IT'S IN THE APPENDIX Crime data covers only the City of LA (91 of 496 neighborhoods), and it adds nothing to price prediction: at the same price level, p = 0.59 and 0.68. Price already carries it.